

Ankit Kumar

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Final-year CS student and Codeforces Expert focused on backend and systems engineering. Passionate about applying algorithmic problem-solving to build high-throughput APIs, storage systems, and performance-oriented infrastructure.

TECHNICAL SKILLS

Languages C++, Java, SQL
Backend Spring Boot, REST APIs, PostgreSQL, WebSockets
Systems Memory allocators, storage engines, Linux systems programming
Infra Docker, AWS EC2, AWS Lambda
Tools Git, Linux, CMake, GDB, perf, TSAN, Google Benchmark

ACHIEVEMENTS

- **TCS CodeVita Season 13** — **Global Rank 15** out of 669,000+ participants
- **Codeforces Expert** (rating 1655) · **LeetCode** — active problem solver across DSA domains

WORK EXPERIENCE

Software Engineer Intern

Feb 2025 – May 2025

ControlShift Talent Pvt. Ltd.

- Built WebRTC signaling server and WebSocket layer with heartbeat, room management, and reconnection for live candidate-AI video sessions
- Offloaded blocking email I/O via async job queues with retry and dead-letter handling; developed 30+ REST endpoints with centralized error handling

PROJECTS

StrataDB — Embedded Key-Value Storage Engine

github.com/ankitr-04/strata-db

C++, CMake, Linux profiling/debugging, perf / TSAN / ASAN / UBSAN / valgrind

- Engineered a NUMA-aware Arena allocator with mmap-backed huge-page fallback and thread-local allocation buffers (TLAB), eliminating global CAS contention on the allocation fast path
- Achieved **~2.0M allocations/sec at 4 threads** with TLABs (**~45,000× fewer CAS operations** vs Arena-only), restoring scalability under multi-core contention
- Designed **O(1)** Arena reset reclaiming 1M nodes at **~22.2M nodes/sec (4.8× faster** than per-node delete); sustained 0.03% TLB miss rate over 100B+ allocations
- Built a lock-free skip-list MemTable with CAS-based insertion, append-versioned MVCC ordering, and flush/stall backpressure; validated with TSAN under 16-thread stress workloads
- Implemented epoch-based memory reclamation with page-aligned 4 KiB per-thread slabs and WAL staging pipeline using CRC32C / XXH3-128 integrity validation with Vyukov MPSC queue handoff

TicketLedger — High-Concurrency Cinema Booking Platform

github.com/ankitr-04/ticket-ledger

Java 21, Spring Boot, PostgreSQL, Stripe

- Enforced zero double-bookings via PostgreSQL SELECT FOR UPDATE with strict lock ordering; analytically bounded hot-seat throughput at **~200 ops/sec** per row from lock-commit latency
- Scaled API layer with Java 21 Virtual Threads: 1M parked connections per JVM at **~1 KB/thread**, isolating HikariCP (20 connections) as the sole bottleneck
- Guaranteed exactly-once payment execution via idempotent Stripe webhook processing with strict financial consistency over UX in late-payment edge cases

Axiom — Redis-Compatible In-Memory Store

github.com/ankitr-04/axiom

Java, NIO, Networking, Concurrent Systems

- Delivered **~81K GET / 68K SET ops/sec** (50 concurrent clients; p99 ≤ 3.1 ms) via a single-threaded NIO event loop with direct ByteBuffer access, eliminating thread contention and GC pressure
- Implemented 29+ Redis-compatible commands: Strings, Lists, Sorted Sets (skip-list), Geospatial (geohash), Streams, and Pub/Sub with pattern matching
- Replicated Redis PSYNC protocol: RDB snapshot on initial sync, async incremental streaming, and WAIT-based sync guarantees; tracked per-replica lag via REPLCONF GETACK

EDUCATION

B.Tech in Computer Science

Aug 2023 – May 2026

Lakshmi Narain College of Technology, Bhopal

CGPA: **8.38 / 10**

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Developer Professional (1Z0-1084-25) — Oracle